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UJIAN AKHIR SEMESTER GASAL 2024/2025
ODD SEMESTER FINAL EXAM 2024/2025

FAKULTAS (Faculty) : KOMUNIKASI DAN INFORMATIKA (Communication And Informatics)			
JURUSAN (Department) : INFORMATIKA (INFORMATICS) (Informatics Engineering)			
Mata Uji - Course	Logika dan Himpunan	Hari/Tanggal - Day/Date	Kamis / 16 Januari 2025
Smt/Kelas - Class	1 / X	Jam ke - Session	1
Penguji - Examiner	1. Prof. Dr. Budi Murtiyasa, M.Kom	Waktu - Duration	90 Menit
Petunjuk - Guidance:			
1. Solve the problems below			
2. Hand written answers			
3. You can use a calculator, not a smartphone			
Capaian Pembelajaran Mata Kuliah - Course Learning Outcomes (CPMK - CLO):			
1.	Able to analyze various computer system methods/algorithms in solving problems in an organization. (50)		
2.	Able to master theoretical concepts in the field of informatics knowledge that are relevant to the needs of multi-platform technology applications. (50)		

Soal Tipe A - Type A Questions

No	Soal - Questions	Nilai - Score	CPMK - CLO
1.	A password must be exactly 6 characters long. Each character can be a digit (0–9) or a lowercase English letter (a–z). How many possible passwords are there if: (a) No restrictions are applied? (b) No character can be repeated? (c) No character can be repeated, must be start and end with e letter?	15	1, 2
2.	Referring to binomial theory, (a) Expand $(2x - y^3)^5$! (b) Find the 3rd term on the expansion $(3p + r^2)^6$! (c) Find the term that contains x^6 on the expansion of $(3x^2 - 2y)^7$!	20	1, 2
3.	A relation $\mathbb{R}$ on a set **A** is called an equivalence relation if it is reflexive, symmetric, and transitive. Let **A** = {1, 2, 3, 4} and $\mathbb{R}$ = {(1,1), (2,2), (3,3), (4,4), (1,2), (2,1)}. Verify is $\mathbb{R}$ an equivalence relation ?.	15	1, 2
4.	Use the principles of mathematical induction to show that: $2 + 4 + 6 + \ldots + 2n = n^2 + n$, for all n natural numbers.	25	1, 2

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